Vishnuvardhan Karudumpa

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Professional Summary

- B.Tech graduate in Computer Science & Engineering (AI & ML) with hands-on experience in NLP, Retrieval-Augmented Generation (RAG), Machine Learning, and Deep Learning. Skilled in Python, TensorFlow, LangChain, FastAPI, and SQL with experience building AI-powered applications.

Education

- **Alliance University** *2022 - 2026*
  Bengaluru, B.Tech in Computer Science and Engineering (AI & ML) CGPA: 7.5/10

Experience

- **Industry Immersion Program** *May 2025 – July 2025*
  - Solved 75+ DSA problems in Python on HackerRank covering arrays, linked lists, trees, and graphs over a 3-month industry immersion program.
  - Practiced 20+ sorting, searching, and recursion-based challenges, improving average problem-solving time by 35% over the program duration.
  - Earned the HackerRank Python certification with a top 15% score, validating proficiency in Python fundamentals, data structures, and algorithmic problem-solving.

Academic Projects

- **Codebase Documentation Search Assistant using RAG** *2026*
  Technologies Used: Python, LangChain, FAISS, FastAPI, BM25 [GitHub] [Demo]
  - Built a search tool using LangChain and FAISS that answers developer queries over 500+ technical documents, cutting resolution time by 60%.
  - Used hybrid BM25 and embedding-based search to find the most relevant results, achieving 87% answer accuracy on test queries.
  - Created a FastAPI interface that returns answers with source references, reducing time spent navigating codebases by 45%.
- **AI Resume Analyzer and Job Match Platform** *2025*
  Technologies Used: Python, NLP, SBERT, SQL [GitHub] [Demo]
  - Built an NLP pipeline using spaCy and SBERT to extract skills and experience from resumes, reducing manual profiling effort by 70%.
  - Scored how well a resume matches a job description using cosine similarity, improving screening accuracy across multiple roles.
  - Stored resume data in a SQL database to support multiple uploads and deliver real-time skill-gap analysis via a web interface.
- **Emotion Recognition System with Smart Recommendations** *2025*
  Technologies Used: Python, TensorFlow, OpenCV, CNN [GitHub] [Demo]
  - Built a CNN model using TensorFlow and OpenCV to detect emotions from face, speech, and text inputs in real time.
  - Applied preprocessing techniques to improve model accuracy, reducing wrong predictions by 30% across varied inputs.
  - Designed a recommendation engine that suggests relevant content based on the detected emotion through a live camera or text interface.

Technical Skills

- **Programming Languages**: Python, SQL
- **AI & ML**: TensorFlow, OpenCV, spaCy, Scikit-learn, NLP, Deep Learning
- **CS Concepts**: DSA, OOP, DBMS, Computer Networks, REST APIs
- **DSA**: Arrays, Linked Lists, Trees, Graphs
- **Tools**: Git, GitHub, Power BI, REST APIs, VS Code, MySQL, Google Colab
- **Languages**: English, Telugu

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional - Oracle *08/2025*
- Python - HackerRank *12/2025*

Achievements and Participations

- Completed a Python coding and placement bootcamp, solving 100+ programming problems. *2025*
- Participated in an Ideathon Competition, contributing ideas for technology-driven problem-solving. *2025*
- Gained hands-on experience in Machine Learning through practical implementation of models. *2023*